CHAPTER 4:

DATA ANALYTIC APPLICATION FOR BUSINESS

(Part 3): Data Investigation & Predictive Analytics

Modern Data Analytics Tool: ALTERYX

Use single workflow for Data Blending, **Analytics** and Reporting.

- A drag and drop visual workflow – no programming required.
- Seamless blending of data.
- 60+ built-in tools for spatial and R-based predictive analytics.
- Simple creation of reports, data for visualization or analytics apps.
- User productivity in hours not days.

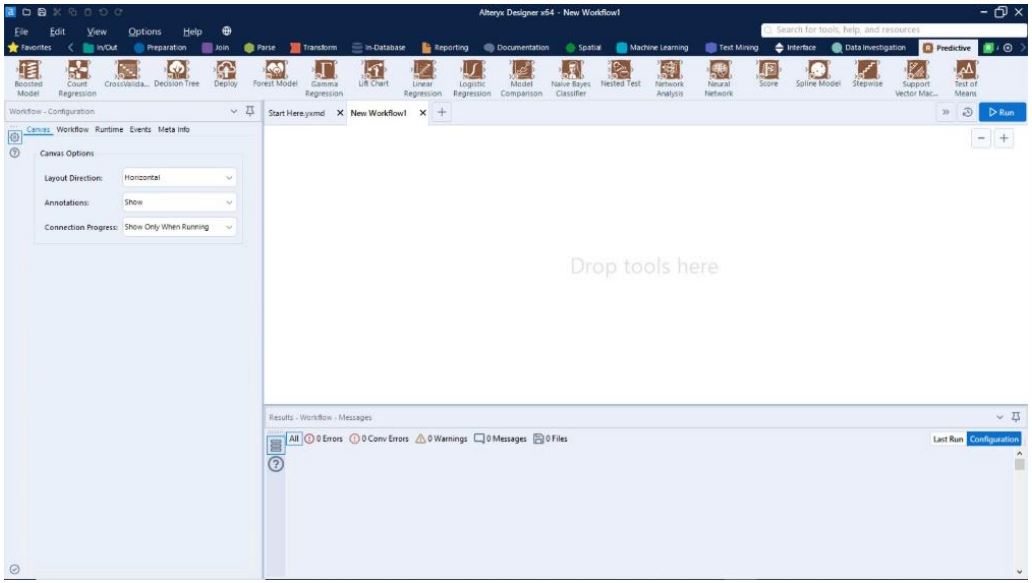
https://www.alteryx.com/



Training - ALTERYX DESIGNER

Tools palette:

- **Data Investigation**
- **Time Series**
- **Predictive**
- **Predictive Grouping**



Data Preparation & Investigation

- The Alteryx predictive tools include many interfaces to help you understand your data in a clean interactive format.

Data Investigation Phase Goals:

1: Investigation

- Discover meaningful and useful data
- Understand significant relationships

2: Preparation

- Clean the data

3: Transformation

- Consolidate, eliminate and transform data for analysis

Data Investigation

- Use Case: **Understanding Restaurant Expenditures**

Question: How can we use this dataset to help us understand the variables that contribute to a household's restaurant expenditures?

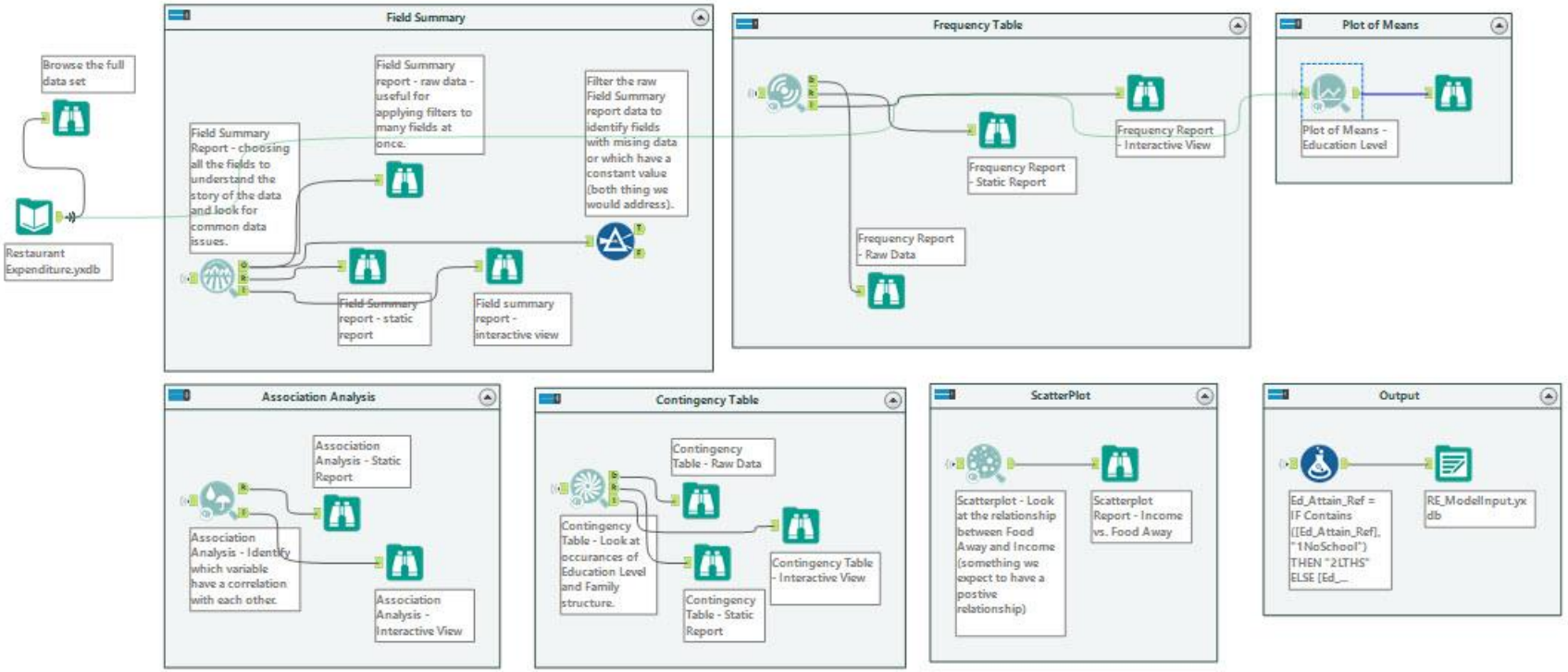
Learn how to:

- ✓ Investigate and clean the data.
- ✓ Understand the relevance and utility of the data.
- ✓ Identify potential predictors and relationships for restaurant.

Data Investigation

Example:

Use a variety of data investigation tools to understand the Restaurant Expenditure dataset



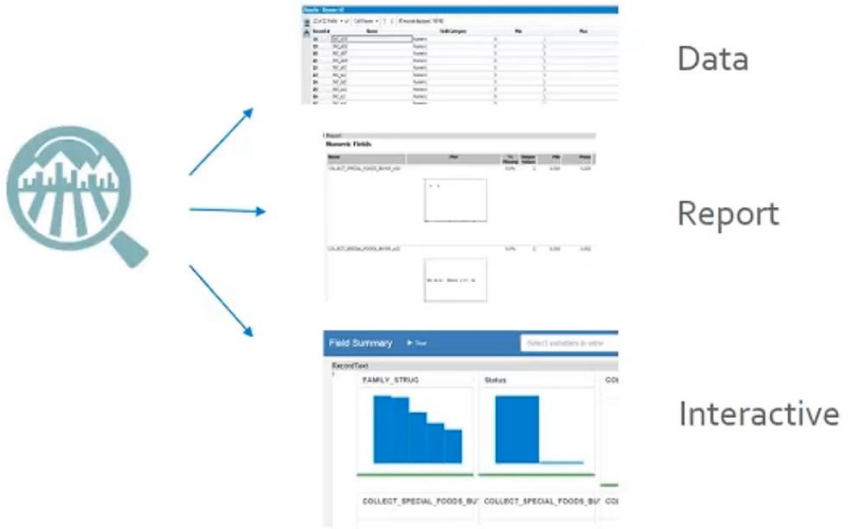
Data Investigation

Data Investigation: Field Summary Tool

Provides info on each field in our data, useful to find:

- Fields with missing values (NULL)
- Constants (fields with same value for every record) Unique Values=1
- Zeros in numeric data? (Min=0)
- Numeric Fields w/ few unique values, is it categorical?
- Values w/ small counts, should they be combined with other categories?

Three Outputs



Data Investigation

Data Investigation: Association Analysis Tool

Association Analysis Results:

- Association measure-strength of the relationship between the 2 variables (-1 to 1)
- P-value – statistical significance
- Correlation matrix

Pearson Correlation Analysis

Focused Analysis on Field Spend

	Association Measure	p-value	
Income	0.505370	0.0000e+00	***
ColDeg	0.265151	0.0000e+00	***
MCwChldrn	0.187470	1.1919e-12	***
AGE_REF	-0.099309	1.8380e-04	***
FAM_SIZE	0.089070	7.9938e-04	***

Full Correlation Matrix

	Spend	Income	AGE_REF	FAM_SIZE	ColDeg	MCwChldrn
Spend	1.000000	0.505370	-0.099309	0.089070	0.265151	0.187470
Income	0.505370	1.000000	-0.132323	0.192446	0.328096	0.273593
AGE_REF	-0.099309	-0.132323	1.000000	-0.328755	-0.106623	-0.321402
FAM_SIZE	0.089070	0.192446	-0.328755	1.000000	-0.056316	0.540003
ColDeg	0.265151	0.328096	-0.106623	-0.056316	1.000000	0.126255
MCwChldrn	0.187470	0.273593	-0.321402	0.540003	0.126255	1.000000

Matrix of Corresponding p-values

	Spend	Income	AGE_REF	FAM_SIZE	ColDeg	MCwChldrn
Spend		0.0000e+00	1.8380e-04	7.9938e-04	0.0000e+00	1.1919e-12
Income	0.0000e+00		5.9344e-07	2.9088e-13	0.0000e+00	0.0000e+00
AGE_REF	1.8380e-04	5.9344e-07		0.0000e+00	5.8872e-05	0.0000e+00
FAM_SIZE	7.9938e-04	2.9088e-13	0.0000e+00		3.4221e-02	0.0000e+00
ColDeg	0.0000e+00	0.0000e+00	5.8872e-05	3.4221e-02		1.9123e-06
MCwChldrn	1.1919e-12	0.0000e+00	0.0000e+00	0.0000e+00	1.9123e-06	

Data Investigation

Data Investigation: Frequency Table Tool

Provides info on each field in our data, useful to find:

- Counts and percentages of each value in a field
- Dig deeper to explore initial findings from Field Summary

Three Outputs

Record #	Field_Name	Field_Value	Frequency	Percent
1	Ed_Attain_Ref	3HSGrad	384	26.93
2	Ed_Attain_Ref	4SomeCol/Assoc	380	26.65
3	Ed_Attain_Ref	5Bachelor's	282	19.78
4	Ed_Attain_Ref	2LTHS	201	14.10
5	Ed_Attain_Ref	6Grad/Prof	177	12.41
6	Ed_Attain_Ref	1NoSchool	2	0.14

Data

Record Table

Ed_Attain_Ref

Value	Frequency	Percent	Cumulative Frequency	Cumulative Percent
3HSGrad	384	26.93	384	26.93
4SomeCol/Assoc	380	26.65	764	53.58
5Bachelor's	282	19.78	1,046	73.35
2LTHS	201	14.10	1,247	87.45
6Grad/Prof	177	12.41	1,424	99.86
1NoSchool	2	0.14	1,426	100.00

Report

Ed_Attain_Ref

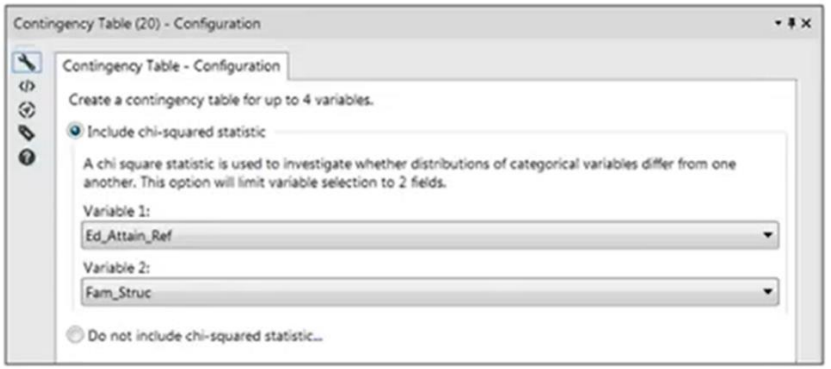


Interactive

Data Investigation

Data Investigation: Contingency Table Tool

- Shows the combination of values between fields by count and percent
- Can be used up to four variables
  - Two for chi-squared statistic



Data Investigation

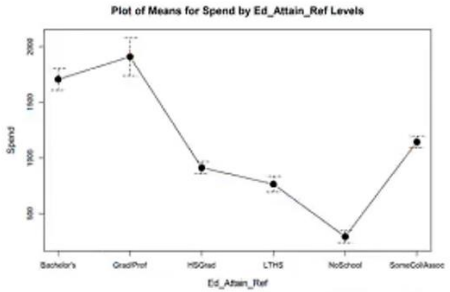
Data Investigation: Plot of Means Tool

Compare two variables:

- Response field: numeric or binary categorical vs. categorical field
- For example:
  - Spend versus educational attainment

One Output





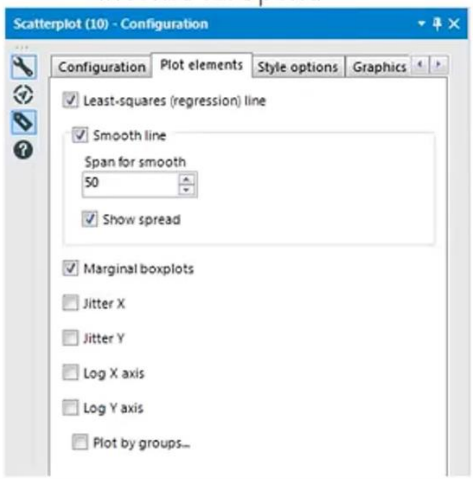
Graph

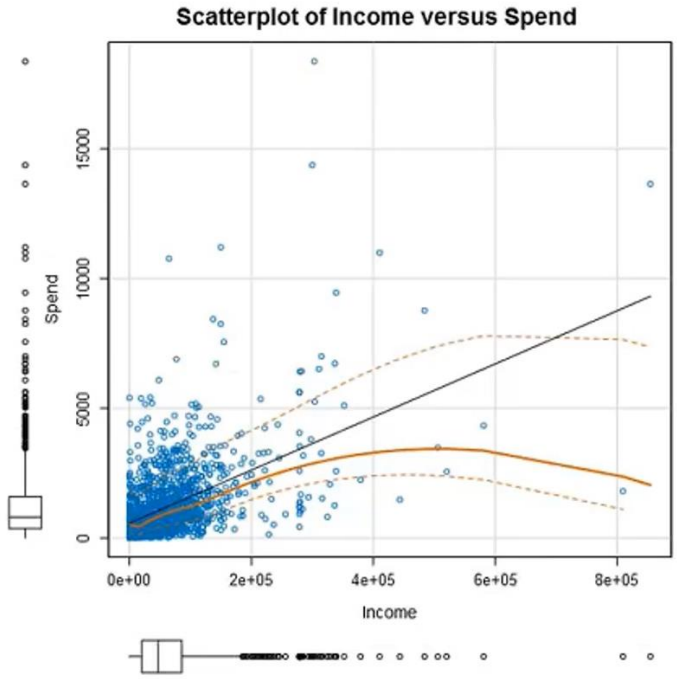
Data Investigation

Data Investigation: Scatterplot Tool

Used to display data for 2 variables:

- Analyze the relationship between the 2 variables
- For example:
  - Income vs. Spend





Time Series Forecasting

- A time series forecasting uses the information in a time series (perhaps with additional information) to forecast future values of that series[2].

- Data required:
  - **Input:** time series data, e.g., monthly sales value

- Time series data often requires cleaning, scaling, and even transformation. For example:
  - ✓ **Frequency** - Perhaps data is provided at a frequency that is too high to model or is unevenly spaced through time requiring resampling for use in some models.
  - ✓ **Outliers** - Perhaps there are corrupt or extreme outlier values that need to be identified and handled.
  - ✓ **Missing** - Perhaps there are gaps or missing data that need to be interpolated or imputed.

- **Performance measures**: Root Mean Squared Error (**RMSE**), Mean Absolute Error (**MAE**), Mean Absolute Percentage Error (**MAPE**), Mean Absolute Scaled Error (**MASE**).

[2] https://machinelearningmastery.com/time-series-forecasting/

Time Series Forecasting

- Time Series analysis accounts for the fact that data points taken over time may have an internal structure (such as autocorrelation, trend or seasonal variation) that should be accounted for.

- There are several methods available to model and forecast time series.

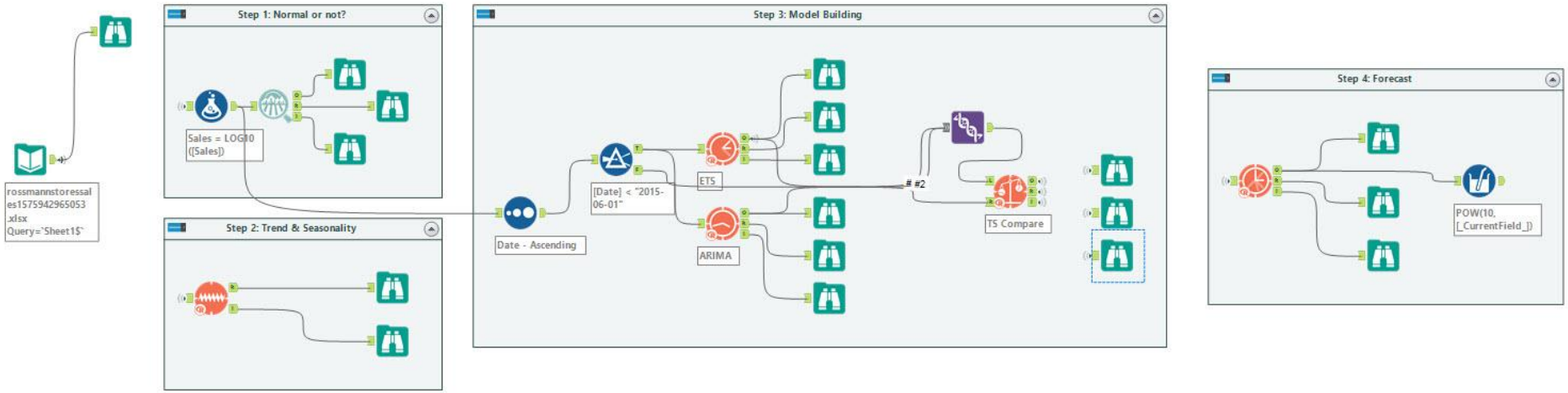
- Learn how to:
  - Use different method between ARIMA and ETS models
  - Forecast time series
  - Compare time series models.

Time Series Modelling

Example:

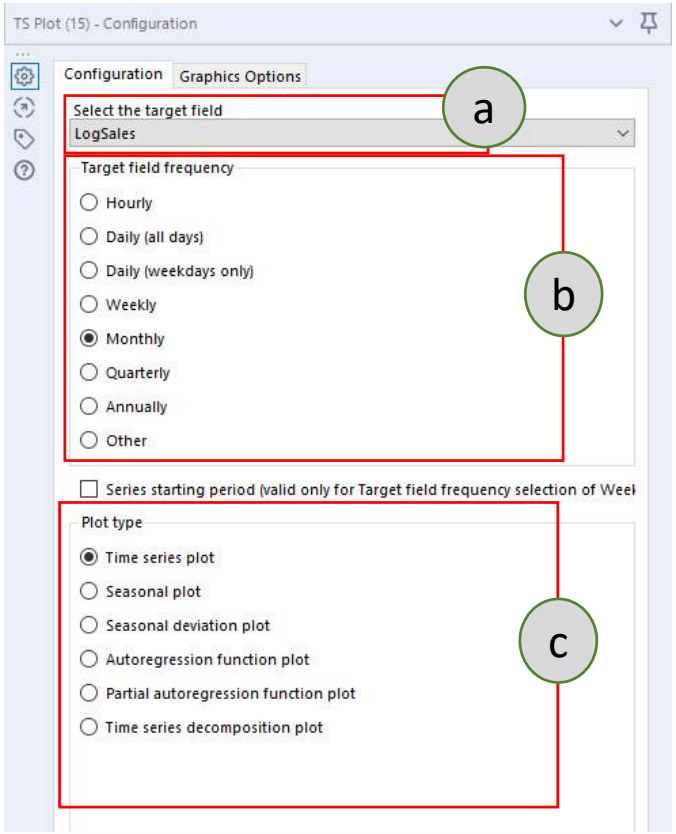
- Use Case: **Forecast the daily sales for RossmanStore**

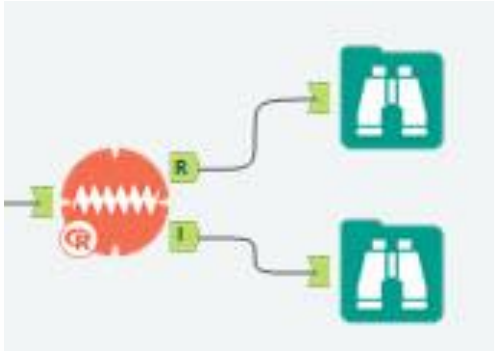
Time Series Forecasting



Time Series Forecasting

- The **TS Plot tool** provides a number of different univariate time series plots that are useful in both better understanding the time series data and determining how to proceed in developing a forecasting model.

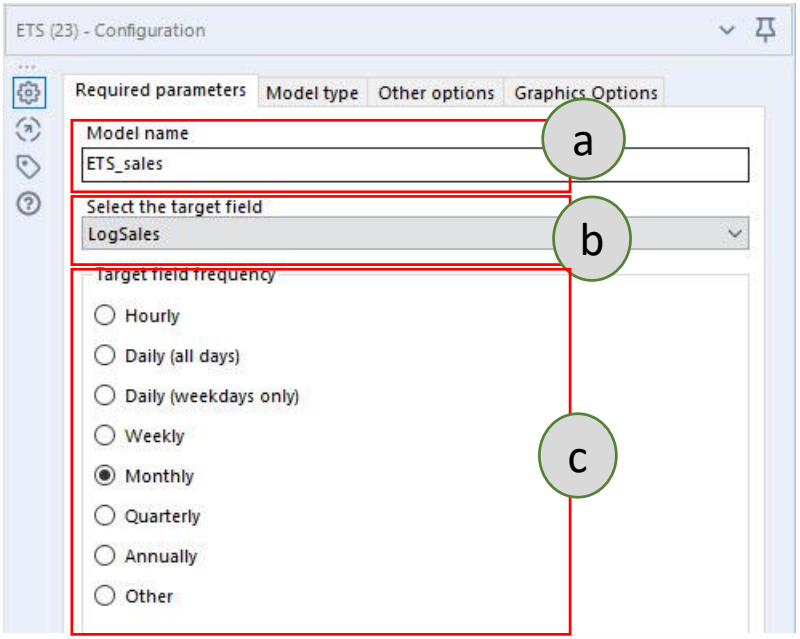


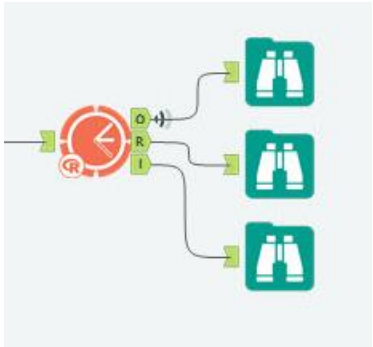


a: Select the field from the data stream for which you wish to create a time series plot.
b: Choose the time interval for the observations of the target field.
c: Select the plot type.

Time Series Forecasting

- Use **ETS tool** to estimate a univariate time series forecasting model using an exponential smoothing method. Exponential smoothing is a commonly used forecasting approach that is based on a weighted average of past observations, with the weights declining in size for more distant past values (the weights are said to follow an exponential decay function).

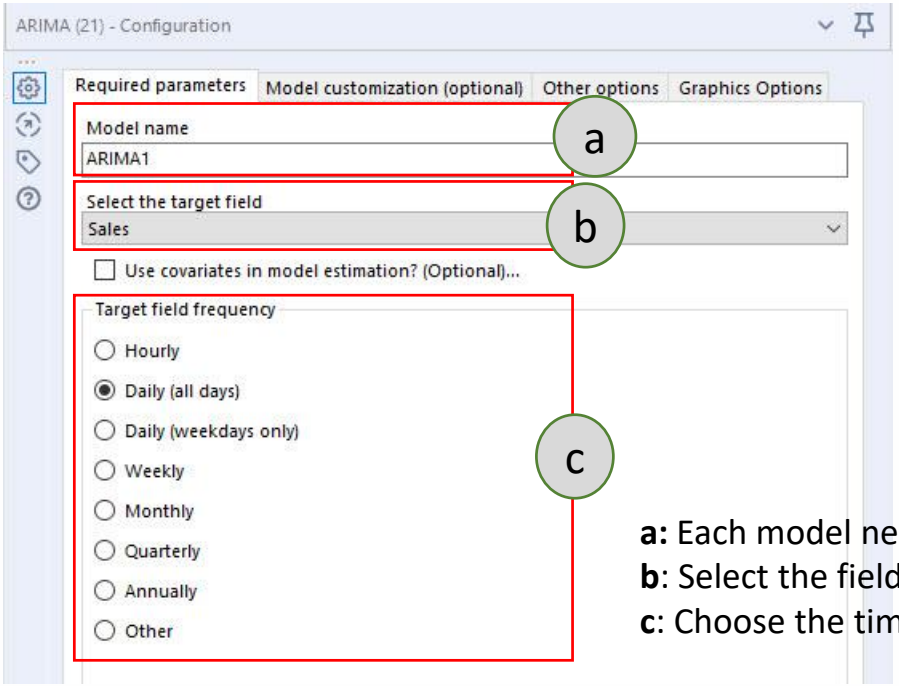


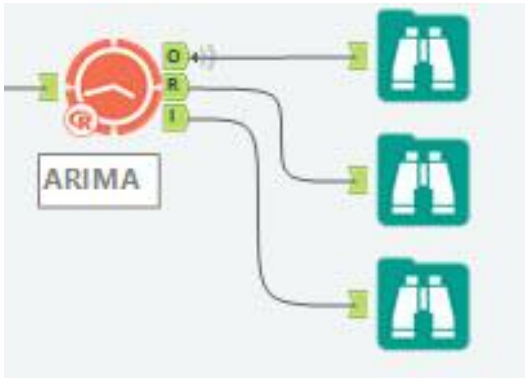


a: Each model needs to be given a name so it can later be identified.
b: Select the field from the data stream you want to forecast.
c: Choose the time interval for the observations of the target field.

Time Series Forecasting

- Use **ARIMA tool** to estimates a time series forecasting model, either as a univariate model or one with covariates (predictors), using an autoregressive integrated moving average (or ARIMA) method. ARIMA is the most commonly used forecasting approach and is considered to be the most general class of models for forecasting a time series field.





a: Each model needs to be given a name so it can later be identified.
b: Select the field from the data stream you want to forecast.
c: Choose the time interval for the observations of the target field.

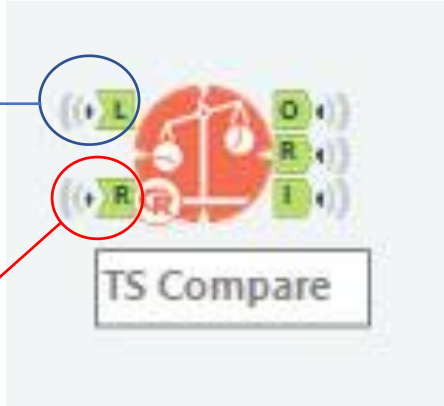
Time Series Forecasting

Use **TS Compare tool** to compare one or more time series models created with either the ARIMA tool or ETS tool, including ARIMA models that use covariates.

This tool provides a number of commonly used measures of model accuracy in terms of comparing each model's point forecasts with the actual values of the field being forecast for a holdout set of data.

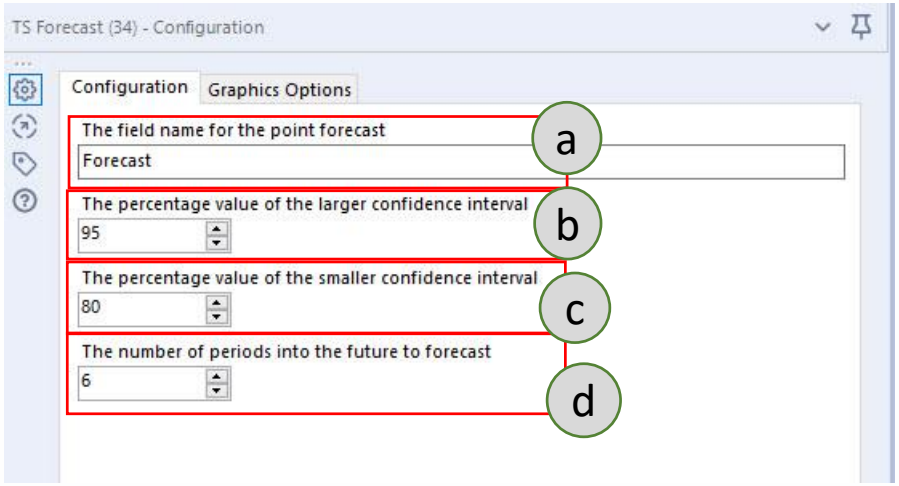
L: connect to a set of time series models that predict the same field, ideally estimating the same time periods, that have been unioned together.

R: Connect to holdout data (10% – 20% from the total samples).



Time Series Forecasting

- The **TS Forecast tool** provides forecasts from a model created with either the ARIMA tool or ETS tool for a user-specified number of future periods. In addition, upper and lower confidence interval bounds are provided for two different (user-specified) percentage confidence levels. For each confidence level, the expected probability that the true value will fall within the provided bounds corresponds to the confidence level percentage.





a: The field name for the point forecasts in the output data stream.
b: Set a value between 1 to 99. Commonly used values for this range are 95 or 99 percent.
c: Set a value between 1 to 99. Commonly used values for this range are 80 or 90 percent.
d: Set the specified number of future periods to be forecast.

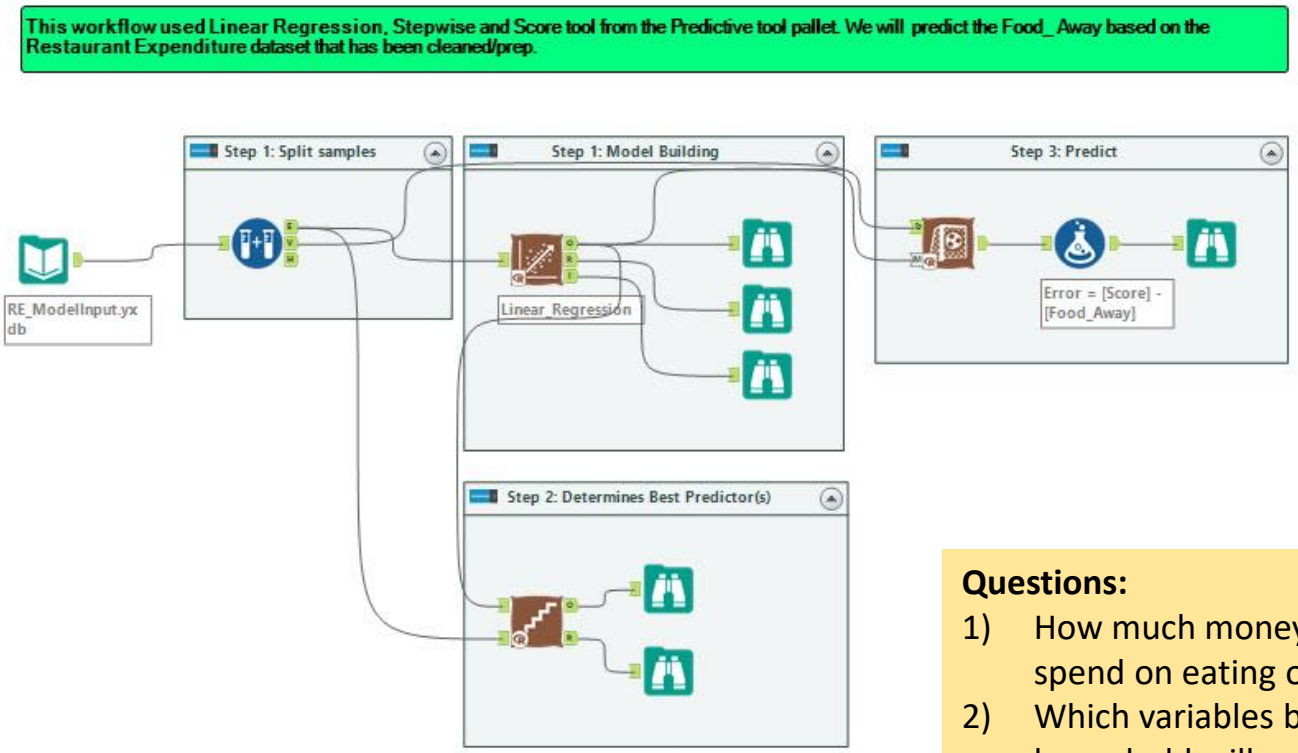
Regression Analysis

- Regression analysis is a form of predictive modelling technique which investigates the relationship between a **dependent** (**target**) and **independent** variable(s) (**predictor**).
- In general, we need to clean and prep the data before we can determine which modelling technique is the best choice for our data.

Regression Analysis

Example:

- Use Case: **Build models that predict Restaurant Expenditures**

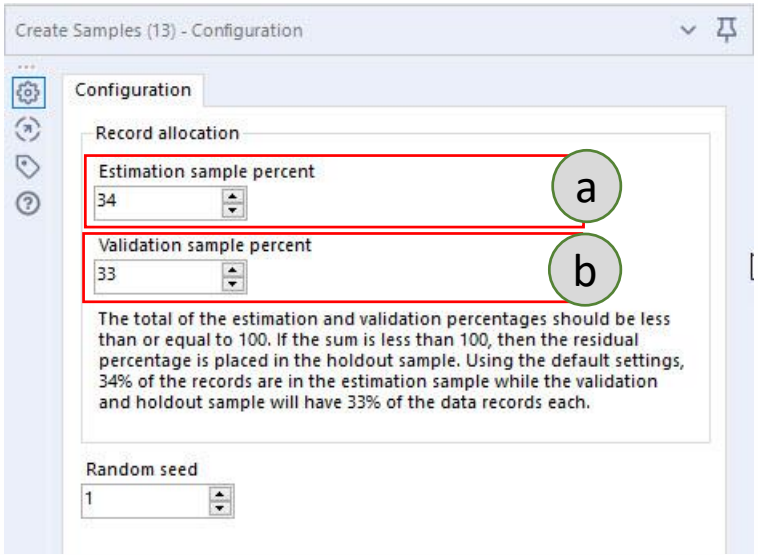


Questions:

1) How much money do we expect each household to spend on eating out?
2) Which variables best indicate how much a household will spend on eating at restaurants?

Regression Analysis

- Use **Create Samples tool** to split the input records into 2 or 3 random samples. In the tool, you specify the percentage of records that are in the estimation (**E**) and validation (**V**) samples. If the total is less than 100%, the remaining records fall in the holdout (**H**) sample.

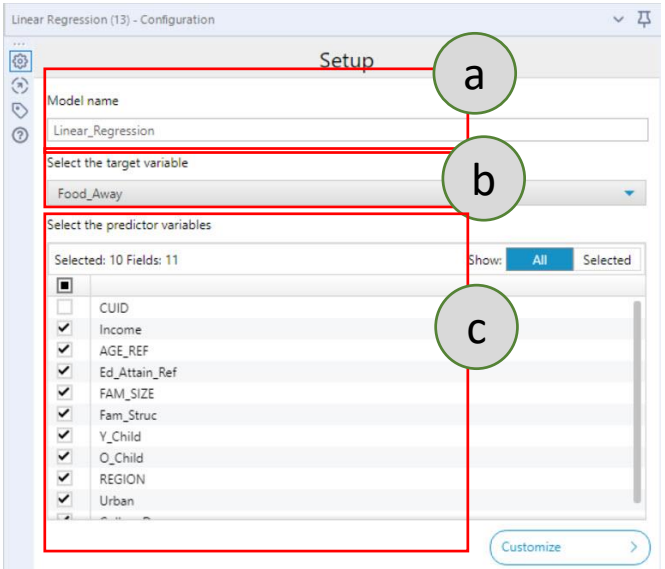


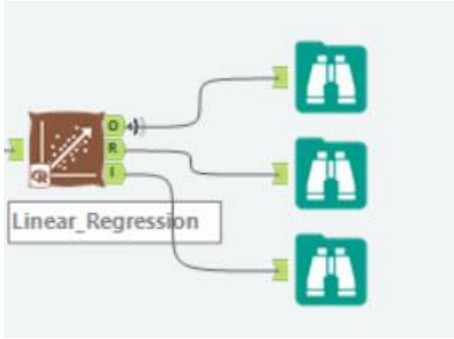


a: specify the percentage of records that are in the estimation (**E**)
b: specify the percentage of records that are in the estimation (**V**)

Regression Analysis

- The **Linear Regression Tool** creates a simple model to estimate values, or evaluate relationships between variables based on a linear relationship.





Output anchor:

O(Output): Display the model name and size of the object in the Result Window

R(Report): Display a summary report of the model that includes a summary and plots.

I(Interactive): Display a dashboard of interactive visualizations to support further data-discovery and model exploration.

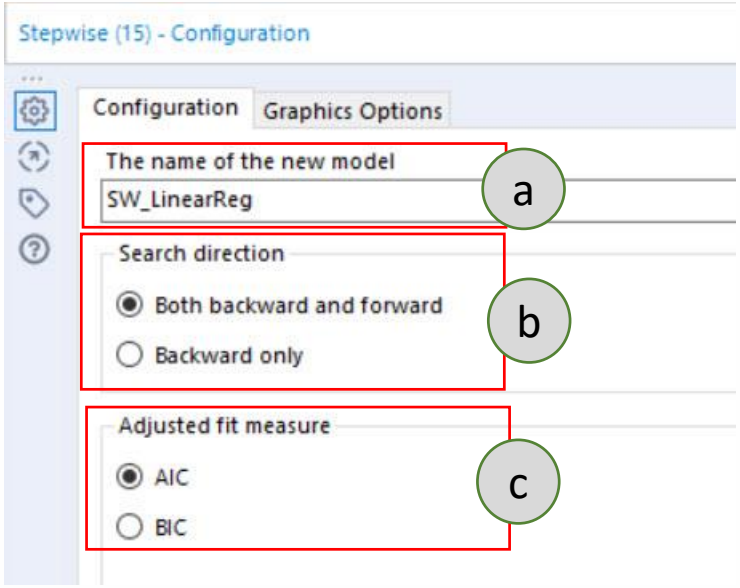
a: Enter a name for the model to identify the model when it is referenced in other tools.

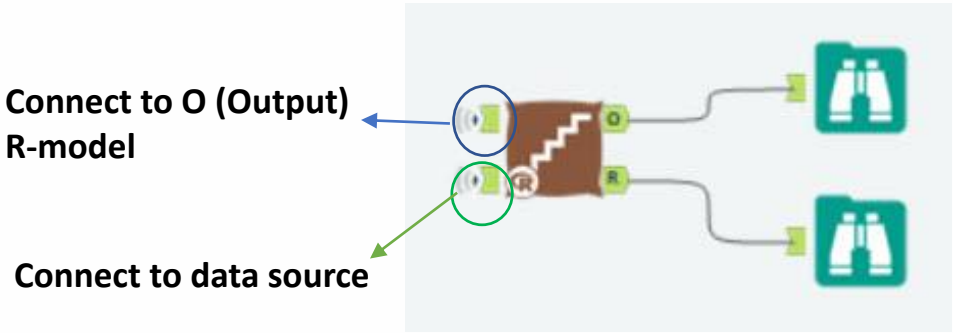
b: Select the data to be predicted.

c: Select the data to use to influence the value of the target variable.

Regression Analysis

- The **Stepwise Tool** determines the best predictor variables to include in a model out of larger set of potential predictor variables for linear, logistic, and other traditional regression models.

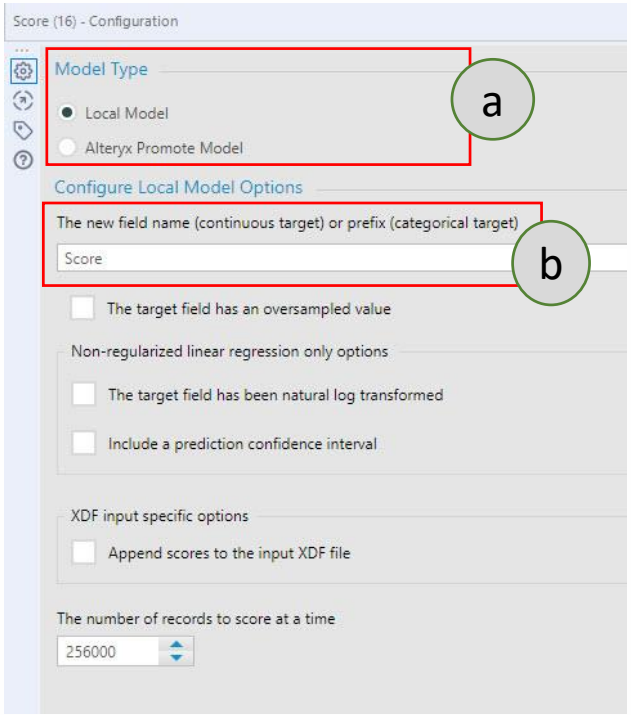


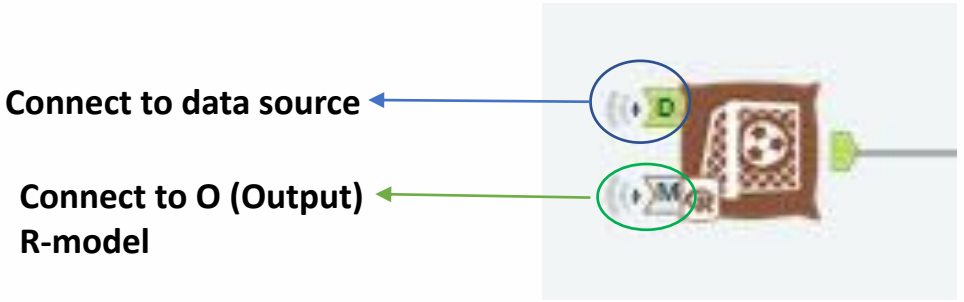


a: Enter a name for the model to identify the model when it is referenced in other tools.
b: Select the method for search direction.
c: Select the fit measure – the criteria used to compare different models and select the best one.
Note: **AIC** – Akaike Information criterion; **BIC** – Bayesian Information criterion. We prefer the model with minimum AIC or BIC value.

Regression Analysis

- The **Score Tool** creates an estimate of a target variable by applying an R-model to a set of supplied predictor variables. If the target variable is categorical, it provides probabilities that a record (based on the predictor variable) belongs to each category. If the target variable is continuous, it estimates the target variable's value. Although it can be used to assess model performance, it does not do so on its own.





a: *Local Model* **-** The model is pulled into the workflow from a local machine or is accessed within a database.
Promote Model - The model is stored in the Promote model management system

b: Write name for the new field.

Classification

- A technique used to predict **categorical** class label.
- Data required:
  - **Input/Features**
  - **Target/Response**
- To prepare the data, the following process might be needed[1]:
  - **Data Cleaning** – to remove the noise and treatment of missing values.
  - **Relevance Analysis** – Database may also have the irrelevant attributes. Correlation analysis is used to know whether any two given attributes are related.
  - **Data Transformation and reduction** – The data can be transformed by any of the following methods.

    Normalization – scale all values for given attribute in order to make them fall within a small specified range. Normalization is used when in the learning step, the neural networks or the methods involving measurements are used.

    Generalization – transform the data by generalizing it to the higher concept. For this purpose we can use the concept hierarchies.

Note – Data can also be reduced by some other methods such as wavelet transformation, binning, histogram analysis, and clustering.

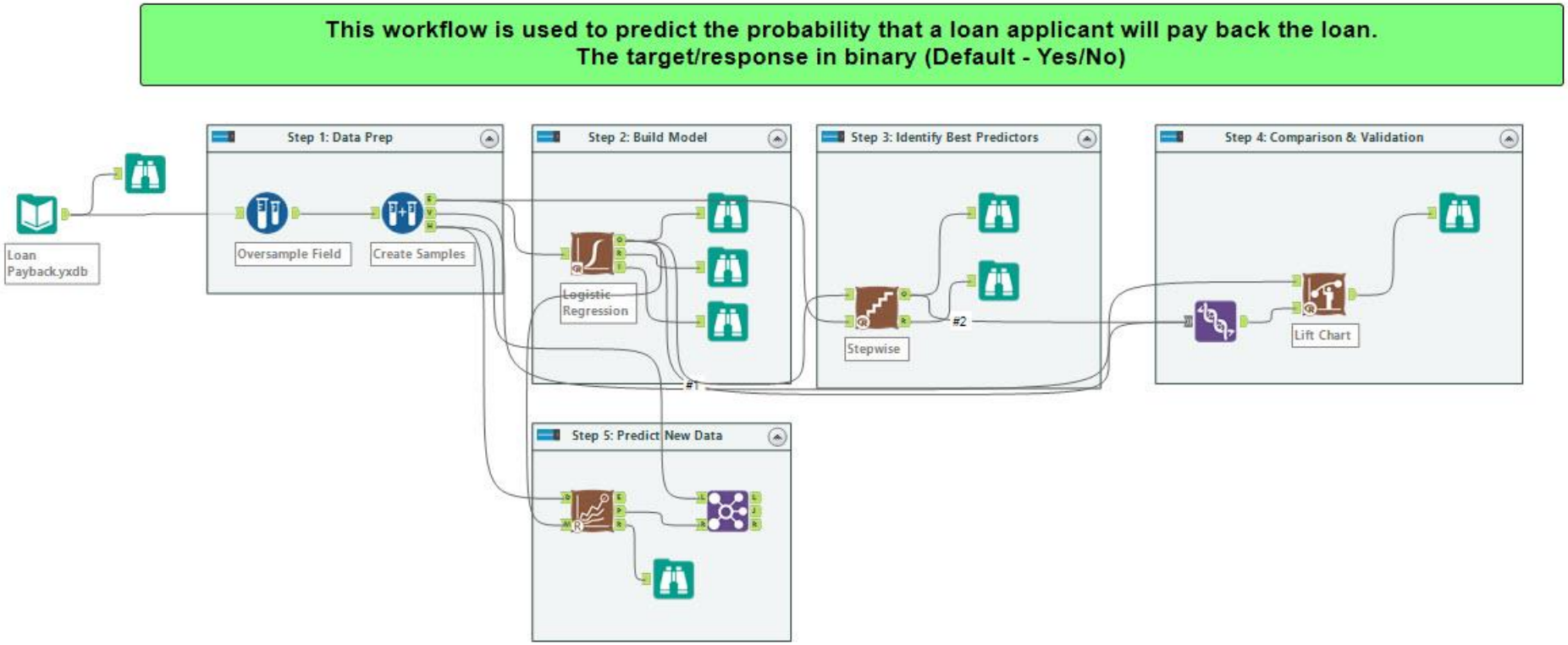
- **Performance measures**: Confusion matrix (**Accuracy**; **F1**; **AUC**)

[1] https://www.tutorialspoint.com/data_mining/dm_classification_prediction.htm

Classification

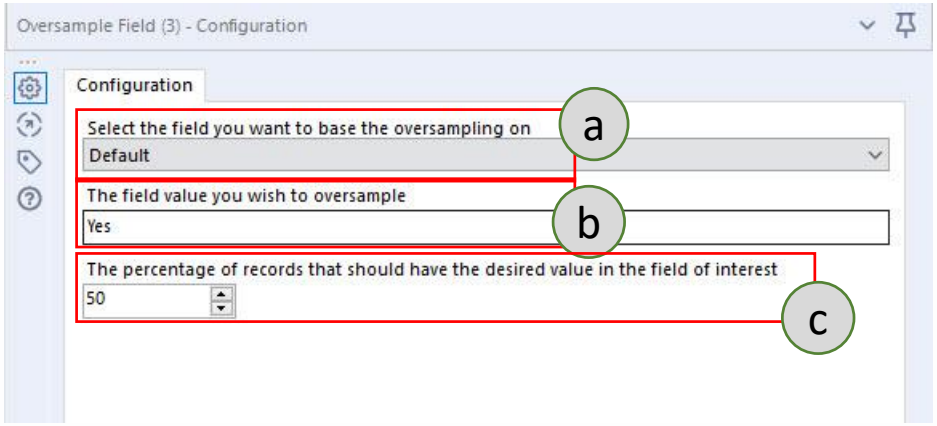
Example:

- Use Case: **Predict Loan Default (binary target)**



Classification

- Use **Oversample Field tool** to create a new sample for analysis that has an elevated percentage of positive responses (often a 50-50 split of positive and negative responses is used).
- It is often the case for data used to develop a binary classification predictive model that the target variable has a much higher proportion of negative (no) responses than positive (yes) responses.





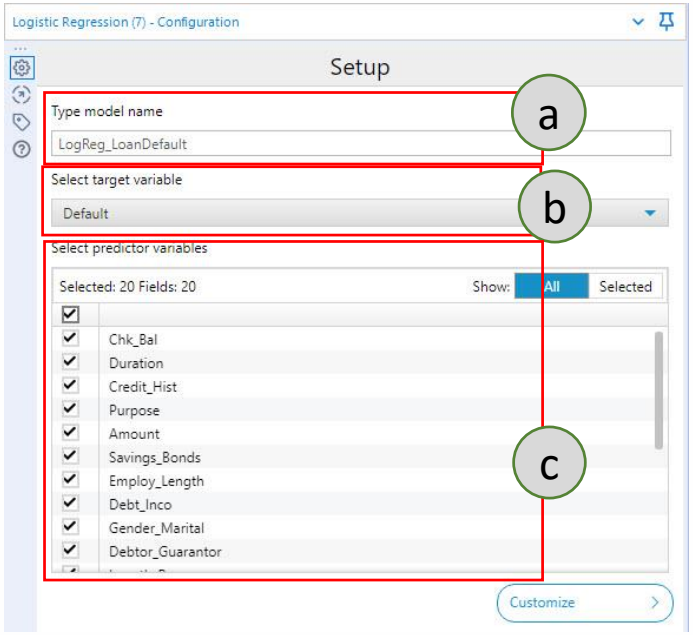
a: The field that contains the value to be oversampled, typically the target variable field in a binary classification predictive model.

b: The level that is to be oversampled, typically the positive ("yes") response in a binary classification predictive model.

c: An integer value between 1 and 100. This value should not be less than the percentage that this level of the field of interest represents in the original data.

Classification

- Use **Logistic Regression tool** creates a model that relates a target binary variable (like a yes/no, pass/fail) to one or more predictor variables to obtain the estimated probability for each of two possible responses for the target variable.

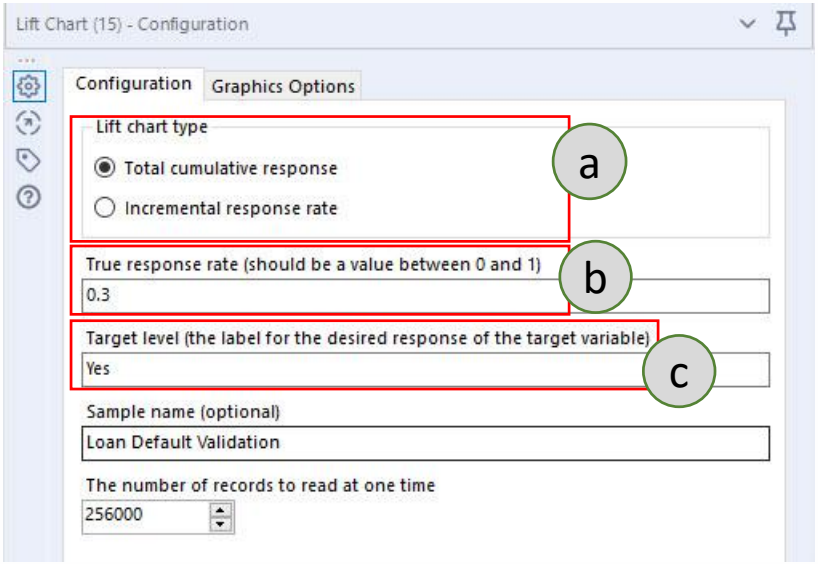




a: A name for the model that can be referenced by other tools.
b: The data field to be predicted, also known as a target/response or dependent variable.
c: The data fields used to influence the value of the target variable, also known as a feature or independent variable

Classification

- The **Lift Chart tool** produces a cumulative captured response chart (also called a gains chart) and an incremental response rate chart that are used to visually assess the comparative accuracy of different binary (yes/no) classification models to predict new data or to make an assessment of the expected economic implications of the use of a predictive model in a business process.





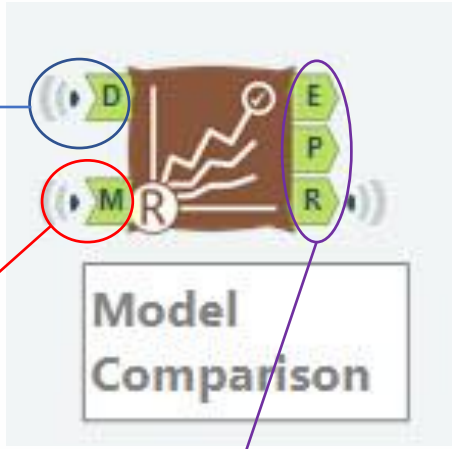
a: Select either Total cumulative response or Incremental response rate.
b: Specify the true (original) response rate of the target variable. Can be obtained from Data Investigation process.
c: Specify target level.

Classification

- The **Model Comparison tool** compares the performance of one or more different predictive models based on the use of a validation, or test dataset. It generates a report, a table of basic error measurements, and a table of prediction results of each model

D: connect to validation dataset (V anchor of Create Samples tool)

M: connect to a set of predictive models that predict the same field, that have been unioned together.



Output anchor:

E anchor: A table of error measures.
P anchor: The actual and the various predicted values.
R anchor: A report containing the error measures and a set of diagnostic plots.

Classification

- Use Case: **Predict the class of Iris plant (multiclass target)**.

- Goal: Find a model that can predict the class of Iris plant with highest accuracy.

Attribute information:

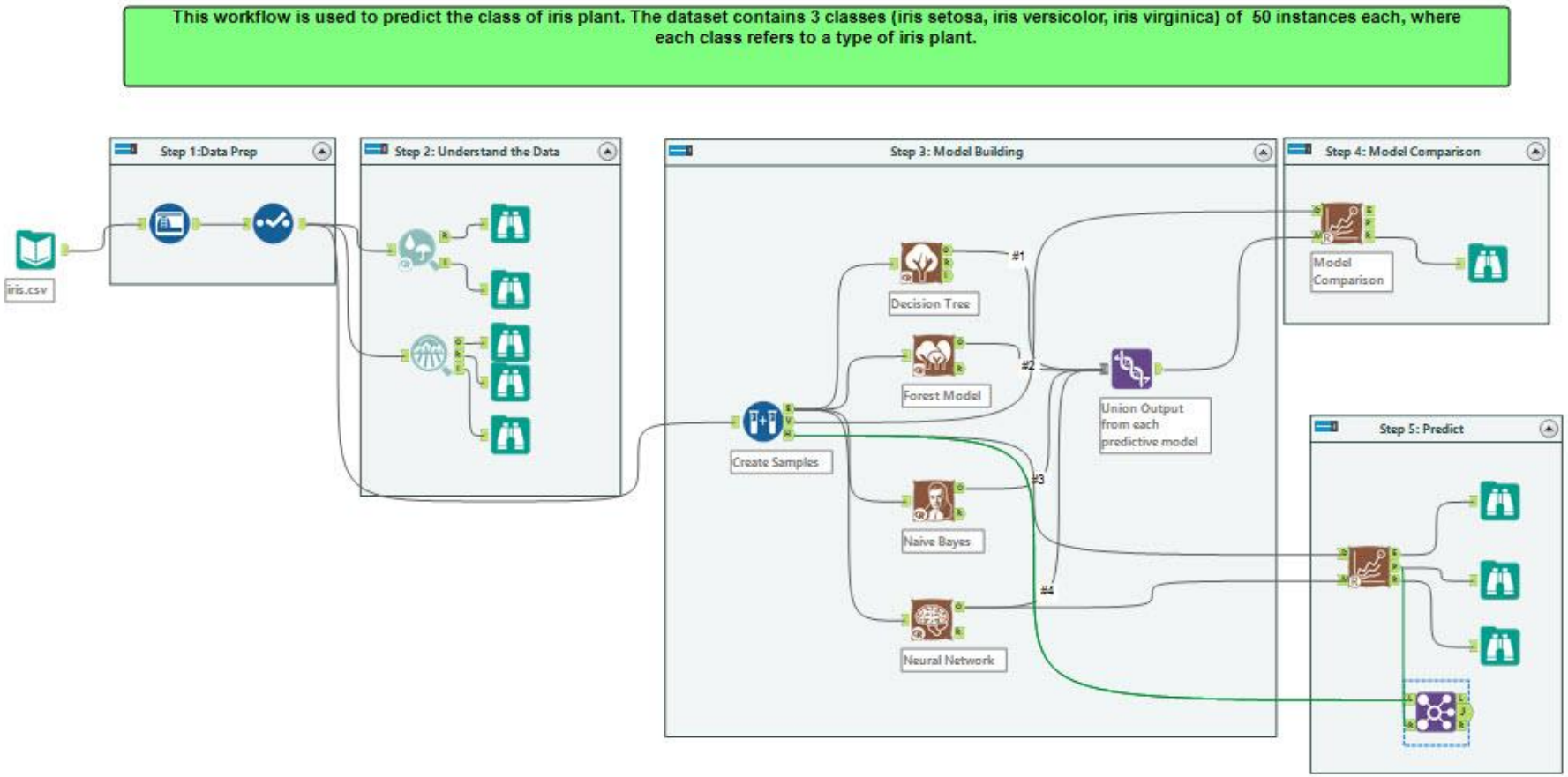
Input/Features:

- ✓ sepal length
- ✓ sepal width
- ✓ petal length
- ✓ petal width

Output/Target: class of iris plant

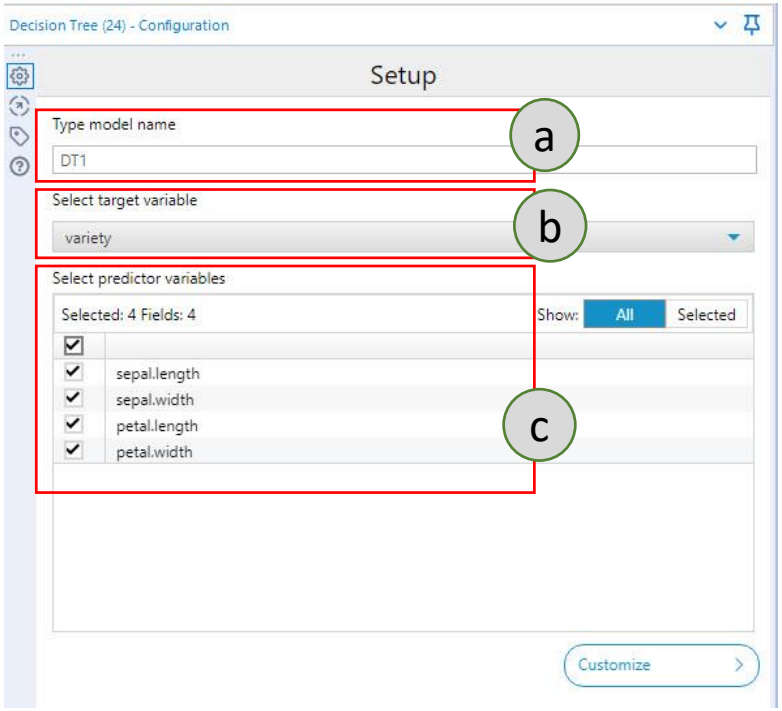
- ✓ Iris Setosa
- ✓ Iris Versicolour
- ✓ Iris Virginica

Classification



Classification

- Use **Decision Tree tool** to create a set of if-then split rules to optimize model creation criteria based on Decision Tree Learning methods.

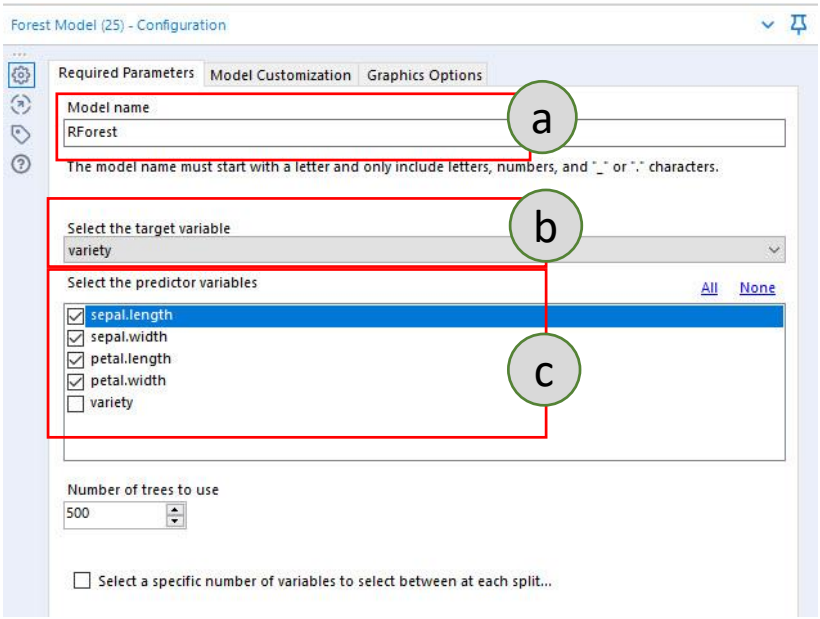




a: A name for the model that can be referenced by other tools.
b: The data field to be predicted, also known as a target/response or dependent variable.
c: The data fields used to influence the value of the target variable, also known as a feature or independent variable

Classification

- The **Forest Model tool** creates a model that constructs a set of decision tree models to predict a target variable based on one or more predictor variables. The different models are constructed using random samples of the original data, a procedure known as bootstrapping. In addition, only a limited number of variables is considered at each tree split, with the number determined set either automatically by R or set by the user.

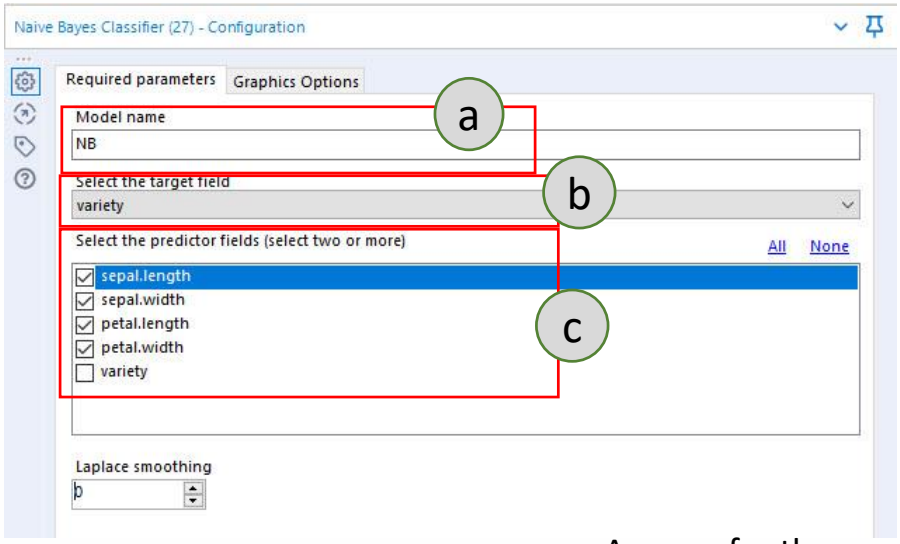




a: A name for the model that can be referenced by other tools.
b: The data field to be predicted, also known as a target/response or dependent variable.
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Classification

- The **Naïve Bayes Classifier tool** creates a binomial or multinomial probabilistic classification model of the relationship between a set of predictor variables and a categorical target variable. The Naive Bayes classifier assumes that all predictor variables are independent of one another and predicts, based on a sample input, a probability distribution over a set of classes, thus calculating the probability of belonging to each class of the target variable.





a: A name for the model that can be referenced by other tools.
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Classification

- The **Neural Network tool** creates a feedforward perceptron neural network model with a single hidden layer. The neurons in the hidden layer use a logistic (also known as a sigmoid) activation function, and the output activation function depends on the nature of the target field.
- Neural networks represent the first machine learning algorithm (as opposed to traditional statistical approaches) for predictive modeling. The motivation behind the method is mimicking the structure of neurons in the brain (hence the method's name).

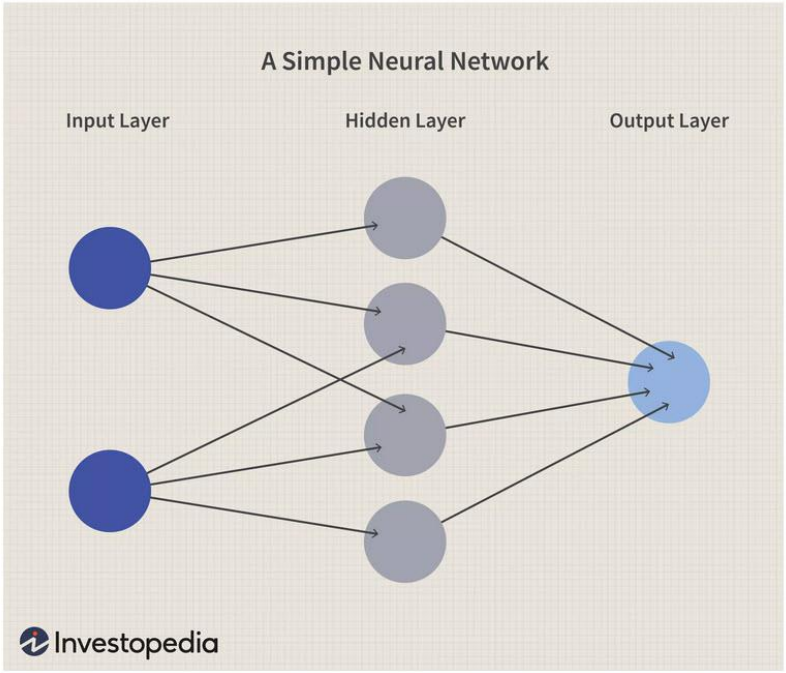
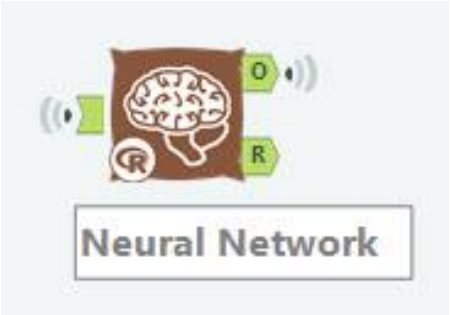
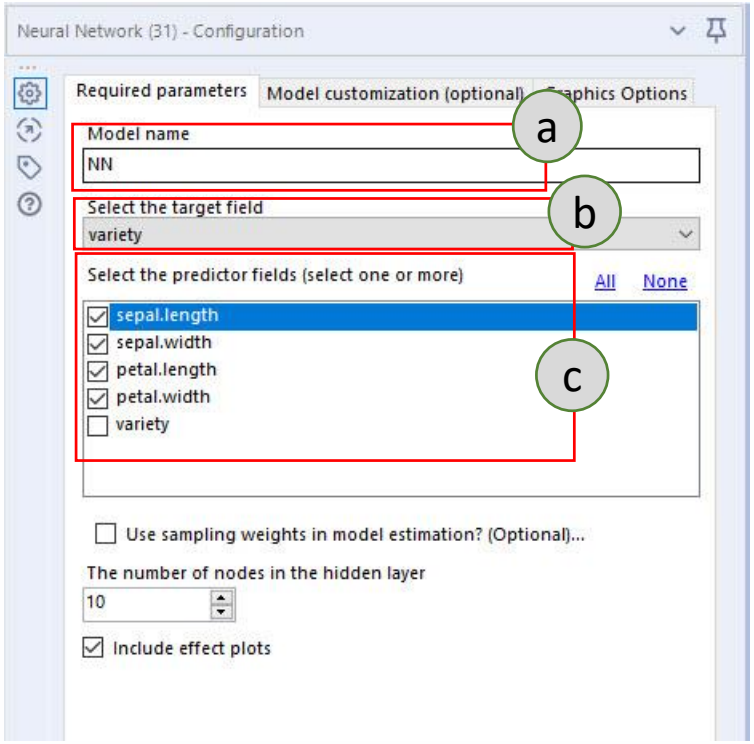
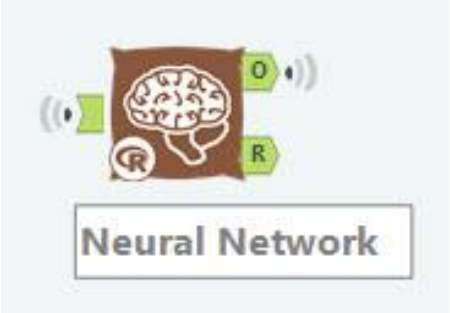


Image by Sabrina Jiang © Investopedia 2020



Classification





a: A name for the model that can be referenced by other tools.
b: The data field to be predicted, also known as a target/response or dependent variable.
c: The data fields used to influence the value of the target variable, also known as a feature or independent variable

Cluster Analysis

- Cluster analysis has a wide variety of use cases, including harnessing spatial data for grouping stores by location, performing customer segmentation or even insurance fraud detection[3].

- Clustering analysis groups individual observations in a way that each group (cluster) contains data that are more similar to one another than the data in other groups.

- Clustering is an unsupervised classification method, which means you do not provide the target groups for analysis.

[3] Tool Mastery/K-Centroid Cluster Analysis/Alteryx Designer Knowledge Base

Cluster Analysis

Use Case: Analyse the restaurant's data to identify the demographic cluster and the corresponding locations.

Questions:

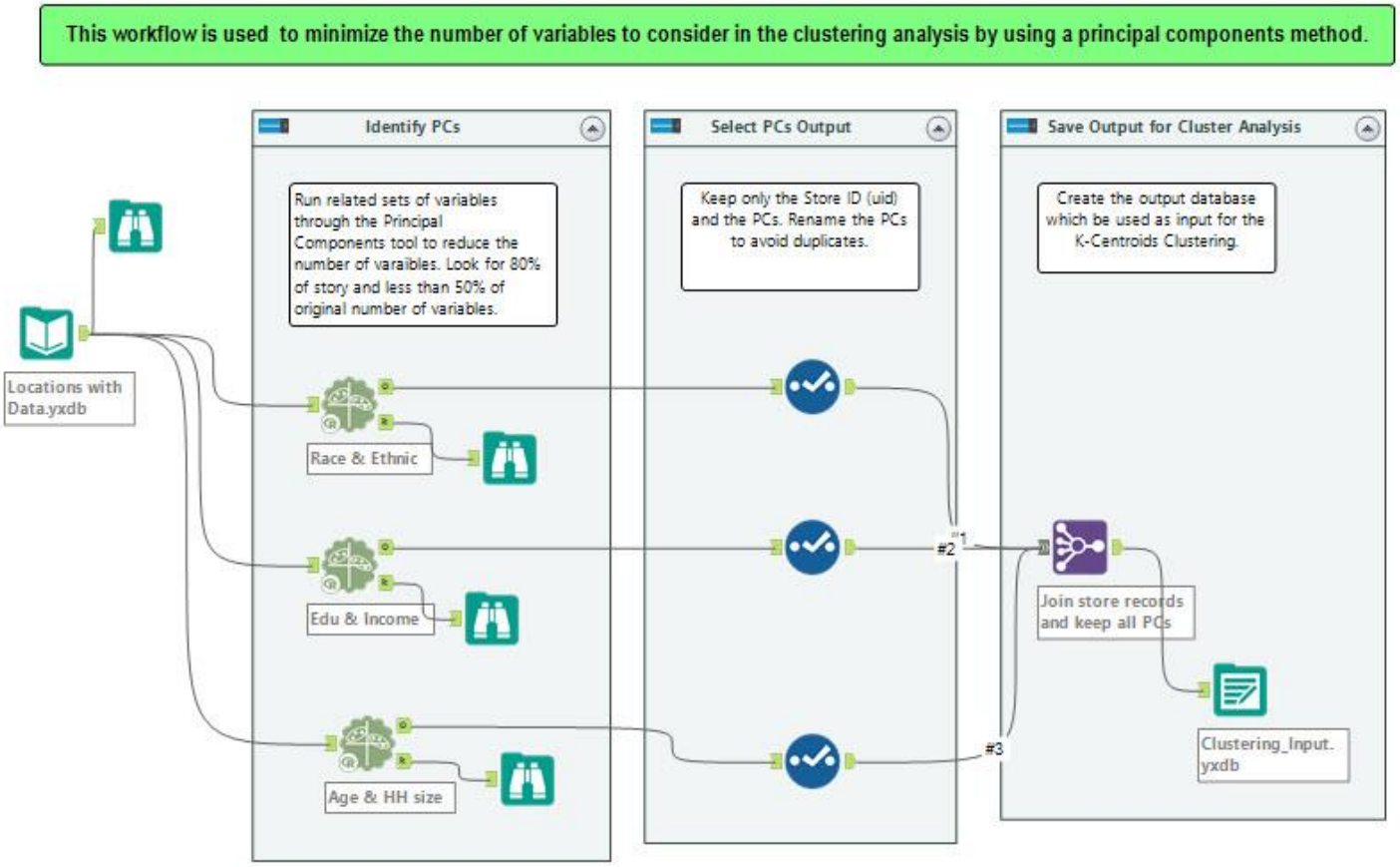
- Which of their customers are similar to other customers?
- Which of their stores has similar demographics in the surrounding areas?

Learn How to:

- Use Principal Components to reduce the number of variables to consider.
- Use K-Means Clustering to determine the number of clusters and which records belong in each.

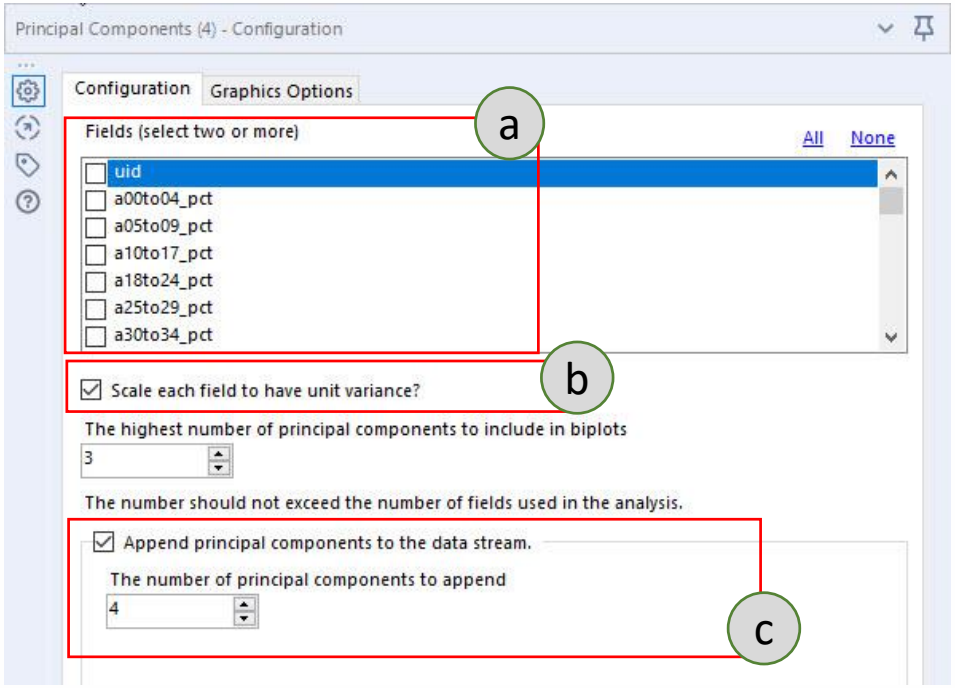
Cluster Analysis – Principal Component

Example:



Cluster Analysis – Principal Component

The Principal Components tool can reduce the dimensions (the number of numeric fields) in a database. It does this by transforming the original set of fields into a smaller set that accounts for most of the variance (i.e., information) in the data. The new fields are called factors, or principal components.





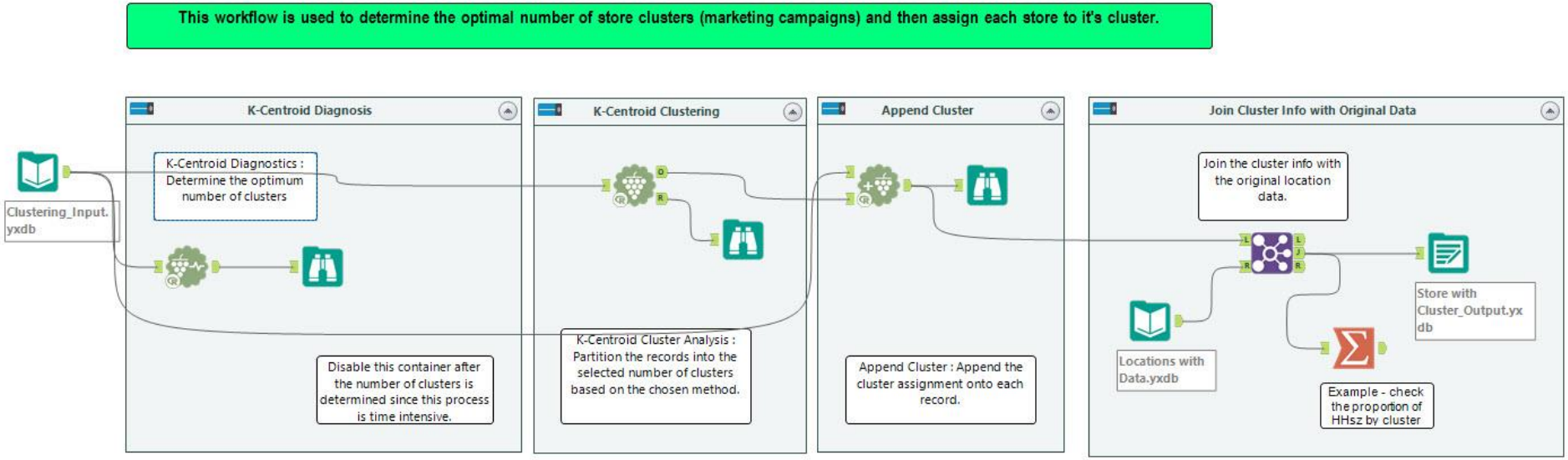
a: Select the numeric fields to use in the principal components analysis.

b: Select this option to standardize the data and use the autocorrelation matrix instead of the autocovariance matrix as a basis for analysis.

c: Select to output the original data along with additional fields for the appended Principal Components. The number of PCs can be adjusted after 1st run.

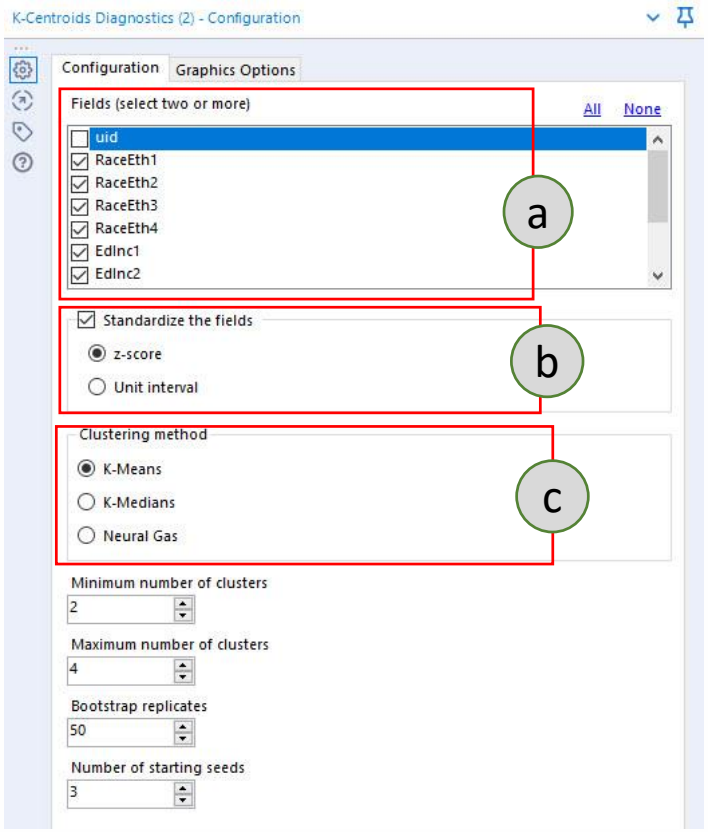
K-Centroids Cluster Analysis

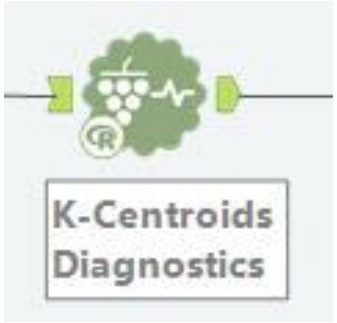
Example:



K-Centroids Cluster Analysis

The **K-Centroids Diagnostic tool** is designed to allow the user to make an assessment of the appropriate number of clusters to specify given the data and the selected clustering algorithm (K-Means, K-Medians, or Neural Gas)





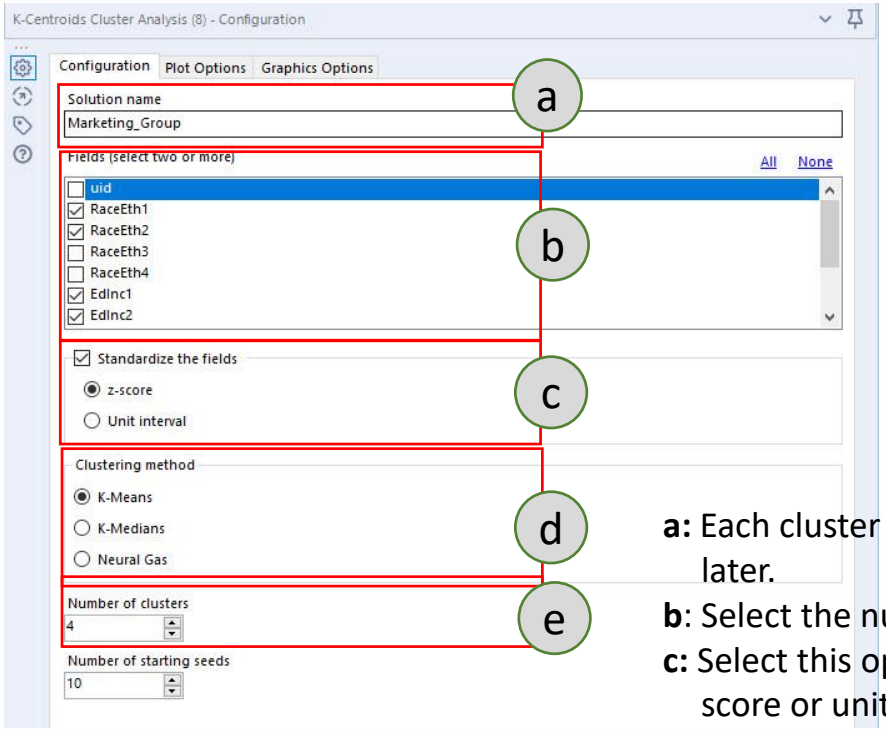
a: Select the numeric fields to be used in constructing the cluster solution.

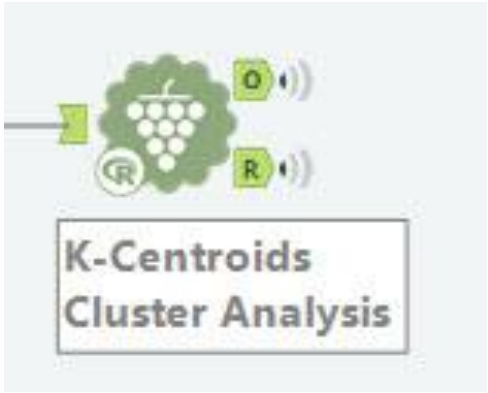
b: Select this option to choose to standardize the variables via either a z-score or unit interval standardization.

c: Choose one of K-Means, K-Medians, or Neural Gas.

K-Centroids Cluster Analysis

- K-Centroids represent a class of algorithms for doing what is known as partitioning cluster analysis.
- These methods work by taking the records in a database and dividing (partitioning) them into the "best" K groups based on some criteria.





a: Each cluster solution needs to be given a name so it can be identified later.

b: Select the numeric fields to use in constructing the cluster solution.

c: Select this option to choose to standardize the variables via either a z-score or unit interval standardization.

d: Choose one of K-Means, K-Medians, or Neural Gas.

e: Set the number according to the result from K-Centroids Diagnostics.

K-Centroid Cluster Analysis

The **Append Cluster tool** appends the cluster assignments from a K-Centroids Cluster Analysis Tool to a data stream.

No configuration required.



Exercises

#1: Forecast the value of sales for Superstore.

Hint: Use the workflow that you have built in lab session (1).

#2: Predict the customer's response for CRM datasets.

Inputs:

- ✓ Consumer
- ✓ Corporate,
- ✓ Home_Office
- ✓ Small_Business

Output/Target: Responder